

Transparency and Decisiveness

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1. Puzzles of modal knowledge

• Decisiveness (D):

For all nonmodal ϕ , $K\phi \vee K\neg\phi \models K\neg(\Diamond\phi \wedge \Diamond\neg\phi)$

- Equivalently: if you know whether ϕ , you know ϕ is settled.
- **Puzzle 1:** An urn contains a mix of black and white marbles only. A marble is selected at random from the urn and placed under a cup. We don't look under the cup. Now Carl looks under the cup without letting anyone else see, and as a result comes to know what color the marble is.
- (1), (2), and (3) seem jointly consistent:
 - (1) It might be black.
 - (2) It might be white.
 - (3) Carl knows whether it's black.
- By (D), (3) entails:
 - (4) Carl knows that (1) and (2) are not both true.
- If knowledge is factive, (4) entails:
 - (5) (1) and (2) are not both true.

ϕ is settled iff $\Diamond\phi$ and $\Diamond\neg\phi$ are not both true, and unsettled iff both $\Diamond\phi$ and $\Diamond\neg\phi$ are true.

Yalcin (2026)

Carl knows whether it's black iff either Carl knows it's black or Carl knows it's white.

• Indecisiveness (I):

For all nonmodal ϕ , $\neg K\neg\phi \wedge \neg K\phi \models K(\Diamond\phi \wedge \Diamond\neg\phi)$

- Equivalently: if you don't know whether ϕ , you know ϕ is unsettled.
- **Puzzle 2:** This time, Alice is an ideal observer just like us. She doesn't look under the cup.
- Putting ourselves into Carl's shoes, (5) and (6) seem consistent.
 - (5) (1) and (2) are not both true.
 - (6) Alice doesn't know whether it's black.
- By (I), (6) entails:
 - (7) Alice knows that (1) and (2) are both true.
- Again, if knowledge is factive, (7) entails:
 - (8) (1) and (2) are both true.

Hawke (2024a, 2024b)

Alice doesn't know whether it's black iff Alice doesn't know it's black and she doesn't know it's white

- It seems that two principles connecting disjunctive knowledge to modal knowledge are individually in tension with a principle governing the use of 'know'.

2. A contextualist response

- **Idea:** at a given context c , '(1) and (2) are not both true' in effect expresses one proposition when unembedded, and a different proposition when embedded under 'Carl knows'.
 - The standard theory:
 - might ϕ : ϕ is true at **some** world in the **relevant domain**.
 - must ϕ : ϕ is true at **every** world in the **relevant domain**.
 - In (4), '(1) and (2) are not both true' expresses a proposition about Carl's evidence, whereas in (5) it expresses a proposition about whatever information is salient in the common ground (e.g., the speaker's evidence).
 - So, although (5) does not follow from (4), this is not a genuine counterexample to factivity.
- However, this response has difficulty predicting our intuitions about (9):
 - (9) ?? It's not raining. Jane knows it might be raining.
 - (10) It's not raining. Jane knows that rain is not ruled out by her evidence.
 - (11) It's not raining. Jane believes it might be raining.

Kratzer (2012)

Ninan (2010), Yalcin (2021).

3. Possible rescue maneuvers

- **Strategy #1:** Anaphoric contextualism
 - A modal can be interpreted anaphorically. The relevant domain for evaluating 'it might be raining' in (9) is restricted by an overt antecedent expression in discourse, i.e. the proposition that it's not raining.
 - According to this view, (12) is the default interpretation of (9):
 - (12) ?? It's not raining. Jane knows that rain is not ruled out by her evidence plus the proposition that it's not raining.
- **Problem:** The mechanisms of anaphoric resolutions are order-sensitive.
 - (13) a. A man walked in. He wasn't wearing a hat.
 - b. ?? He wasn't wearing a hat. A man walked in.
 - (14) a. ?? It's not raining. Jane knows it might be raining.
 - b. ?? Jane knows it might be raining. It's not raining.

Stojnić (2017)

I think this view faces another challenge. If the denotation of 'might' is restricted by an anaphorically retrieved information in (9), why is the same restriction unavailable in (11)?

- **Strategy #2:** Presuppositional contextualism

Perl (2017)

- The epistemic use of a modal presupposes that the **relevant domain** is *epistemically correct*, i.e. it contains all truths.
- According to this view, (9) sounds infelicitous because it carries a presupposition that contradicts the asserted content:

(9) It's not raining. Jane knows it might be raining.

Asserted content: It's not raining. Jane knows that rain is compatible with Jane's evidence.

Presupposition: Jane's evidence is epistemically correct.¹

¹ More precisely, (9) presupposes that Jane's evidence is compatible with rain and is epistemically correct.

- **Problem #1:** On this view, the appropriate attitude toward a presupposition must be acceptance. But if the norm of assertion is belief, or any other attitude stronger than acceptance, it's hard to see why (9) sounds so bad. After all, one can accept something for the sake of argument/conversation/inquiry while knowing it to be false.
- **Problem #2:** It's hard to see how this view can explain the felicity of (15) without invalidating some classically valid principles.

(15) It might be raining and it might not be raining. But S knows whether it's raining (i.e., either S knows that it's raining or S knows that it isn't raining).

Beddor and Goldstein (2021).

(16) ?? Either (it might be raining and it might not be raining and S knows it's raining) or (it might be raining and it might not be raining and S knows it isn't raining).

4. A new approach

4.1. Knowledge ascriptions

- We start with the following observations:

(17) a. ?? Escargot is tasty. But it isn't tasty.

b. ?? Escargot is tasty. But S knows it isn't tasty.

(18) a. ?? It isn't raining. But it might be raining.

b. ?? It isn't raining. But S knows it might be raining.

Consider (17b). Even if it's contextually salient that S loathes snails, it's difficult to interpret each occurrence of 'tasty' in (17b) relative to distinct standards of taste.

The same goes for (18b). Even if it is contextually salient that S's knowledge is compatible with rain, (18b) still resists a good interpretation.

In syntactic configurations like (17b) and (18b), an intersentential context-shift is hard to obtain. We are strongly inclined to interpret the first conjunct and the complement clause of the second conjunct as making conflicting claims about the same situation.

- (18b) is infelicitous not because S fails to know it might be raining when it isn't raining in the actual world, but because S's knowledge can't be reported with such a that-clause in syntactic configurations like (18b).

Further evidence comes from the contrast between (15) and (19):

(15) It might be raining and it might not be raining. But S knows whether it's raining.

(19) ?? It might be raining and it might not be raining. But S knows it's false that it might be raining and it might not be raining.

Go back to the marble case:

(20) It might be black and it might be white. But Carl knows what color the marble is.

(21) ?? It might be black and it might be white. But Carl knows it's false that it might be black and it might be white.

4.2. Restricted transparency theory

- We start with a simple idea:

Transparency theory: 'An agent x knows that might ϕ ' is true iff x 's knowledge state leaves open ϕ .

- Traditionally, x 's knowledge state is represented as a set of possible worlds compatible with the **content** of what x knows.
- Accepting this theory has unhappy consequences. For example,
 1. It validates **Knowledge Indecisiveness (KI)**: $\neg K\neg\phi \models K\Diamond\phi$
 2. Given factivity, it validates **Modal Omniscience (MO)**: $\phi \models K\Diamond\phi$
- I think (KT) and (MO) ought to be rejected. But I don't think we should abandon the entire transparency theory. I am going to defend a restricted transparency theory of modal knowledge.
- My working hypothesis:

Hypothesis: Modal verbs project their domains from a part of the evaluation world – their 'modal anchors'.

Take 'must'. In (22), a modal anchor could be a situation where someone comes in from the street with wet clothes. The relevant domain is represented as a set of worlds that have an exact match of that anchor. (22) is true just when it's raining at every world in that domain.

(22) It must be raining.

- The semantics for 'might' goes like this:

See Hacquad (2010), Arregui (2009), Kratzer (2012, 2020), Alonso-Ovalle and Menéndez-Benito (2017).

Factual domain projection: for any part of the world s , $f(s)$ is a set of worlds that have an exact match of s .

'might ϕ ' is true at w iff there exists a situation s (modal anchor) such that

- (i) s is a part of w , $s < w$, and
- (ii) some worlds $w' \in f(s)$ contain a situation s' such that (iia) s' is an exact match of s , and (iib) ϕ is true at s' .²

- To build up an anchor semantics for mighty knowledge, I need two further hypotheses:

Anchoring requirement: the verb 'know' in sentences of the form 'x knows that might ϕ ' is invariably anchored to the anchoring situation for 'might' *with which x is properly acquainted*.

Boundedness requirement: the modal domain projected from the anchor for 'might' is always a subset of the epistemic domain projected from the anchor for 'know'.

The anchoring requirement guarantees that the epistemic domain for 'know' be a subset of the modal domain for 'might', since mereologically speaking, each world that includes an anchoring situation for 'might' with which x is properly acquainted must be a world that includes an anchoring situation for 'might'.

Together with the boundedness requirement, we have:

Restricted transparency theory:

Given that there exists an anchor for 'know' at w , 'an agent x knows that might ϕ ' is true at w iff x 's knowledge state at w leaves open ϕ .

- Where s is the modal anchor for 'might',
'an agent x knows that might ϕ ' is true at w iff there is an acquaintance relation A such that
 - (i) x bears A uniquely to s at w , $A(w, x) = s$, and
 - (ii) some worlds $w' \in f(A(w, x))$ contains a situation $A(w', x')$ such that $A(w', x')$ is an exact match of $A(w, x)$, and ϕ is true at $A(w', x')$.

5. Consequences

- **Factivity:** $K\phi \models \phi$
 - Consider (9):
 - (9) ?? It isn't raining. Jane knows it might be raining.

² For simplicity, I assume that the anchoring situation is identical to the evaluation situation. But in principle these two situations should be connected via a contextually supplied relation R , which is not necessarily an identity relation.

For an account of the acquaintance relation, see, among others, Lewis (1983).

the boundedness requirement might be located at the level of presupposition. I'm open to this possibility. For a similar constraint, see Mandelkern (2024).

- For the first conjunct and the modal prejacent of the second conjunct are read as making claims about the same situation, the situation where it isn't raining also serves as the modal anchor for 'might'.
- On this account, (9) describes a metaphysical impossibility. For (9) to be true, some worlds in Jane's knowledge state must be such that it's raining in a situation that is an exact match of the situation where it isn't raining.

- **Indecisiveness:** $\neg K\neg\phi \wedge \neg K\phi \models K(\Diamond\phi \wedge \Diamond\neg\phi)$

Decisiveness: $K\phi \vee K\neg\phi \models K\neg(\Diamond\phi \wedge \Diamond\neg\phi)$

Modal Omniscience: $\phi \models K\Diamond\phi$

- Our theory validates (D) but not (I).

The validity of (D) is straightforward.

Fake letter: Alice receives a letter from a diligent lab assistant correctly informing her that she probably finds her friend Bert attractive, even though most such letters being sent out by disgruntled assistants are fake. And in fact, Alice does find Bert attractive. Reading the letter, she comes to have high credence that she finds Bert attractive.

adapted from Moss (2018).

(23) ✓ Alice doesn't know that Bert is unattractive.

(24) × Alice knows that Bert might be attractive.

- (24) can be true only if Alice is properly acquainted with the fact that the letter was sent out by a diligent lab assistant. Since this condition doesn't satisfy, (24) isn't true.
- Holding fixed the same anchor for 'know' in both (23) and (24), we get a counterexample to (I).

- **Knowledge Entails Belief (KEB):** $K\phi \models B\phi$

- Our theory does not validate (KEB).
- Imagine a case where the appropriate way to be acquainted with the anchor for the knowledge prejacent is causal, rather than doxastic. 'x knows that p' is true but 'x believes that p' is false.

For an interesting case for this point, see Holguín (2021).

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